UVA CS 4774: Machine Learning

S0: Lecture 00: Weekly Quiz and QA

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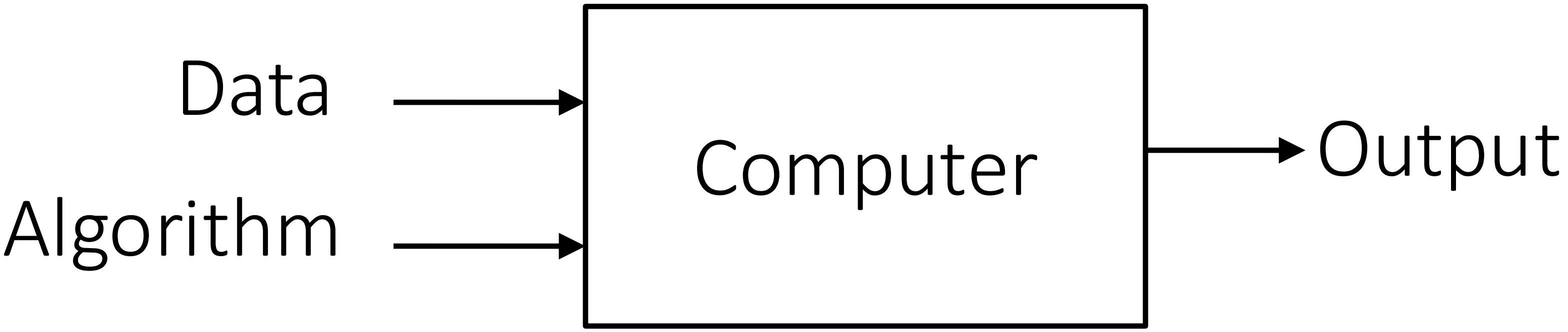
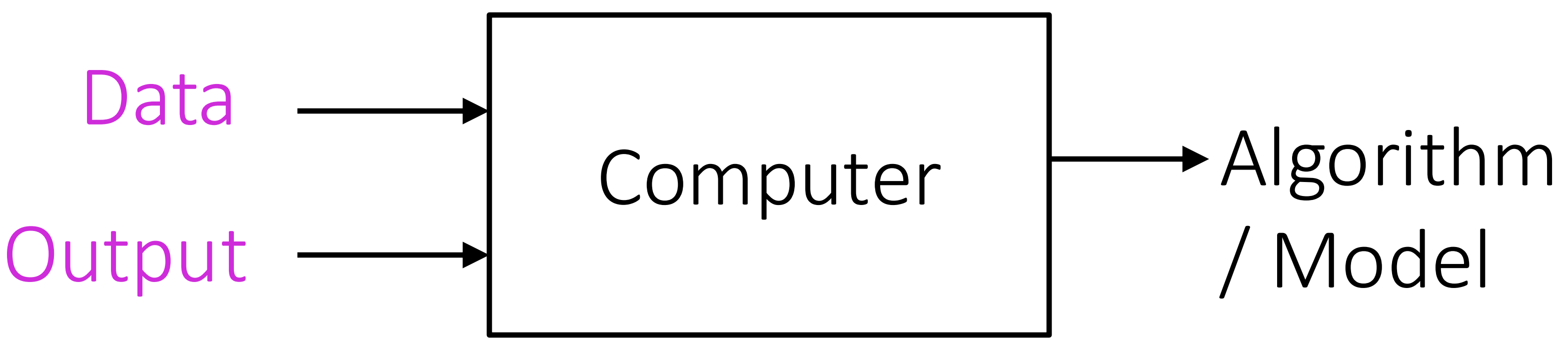
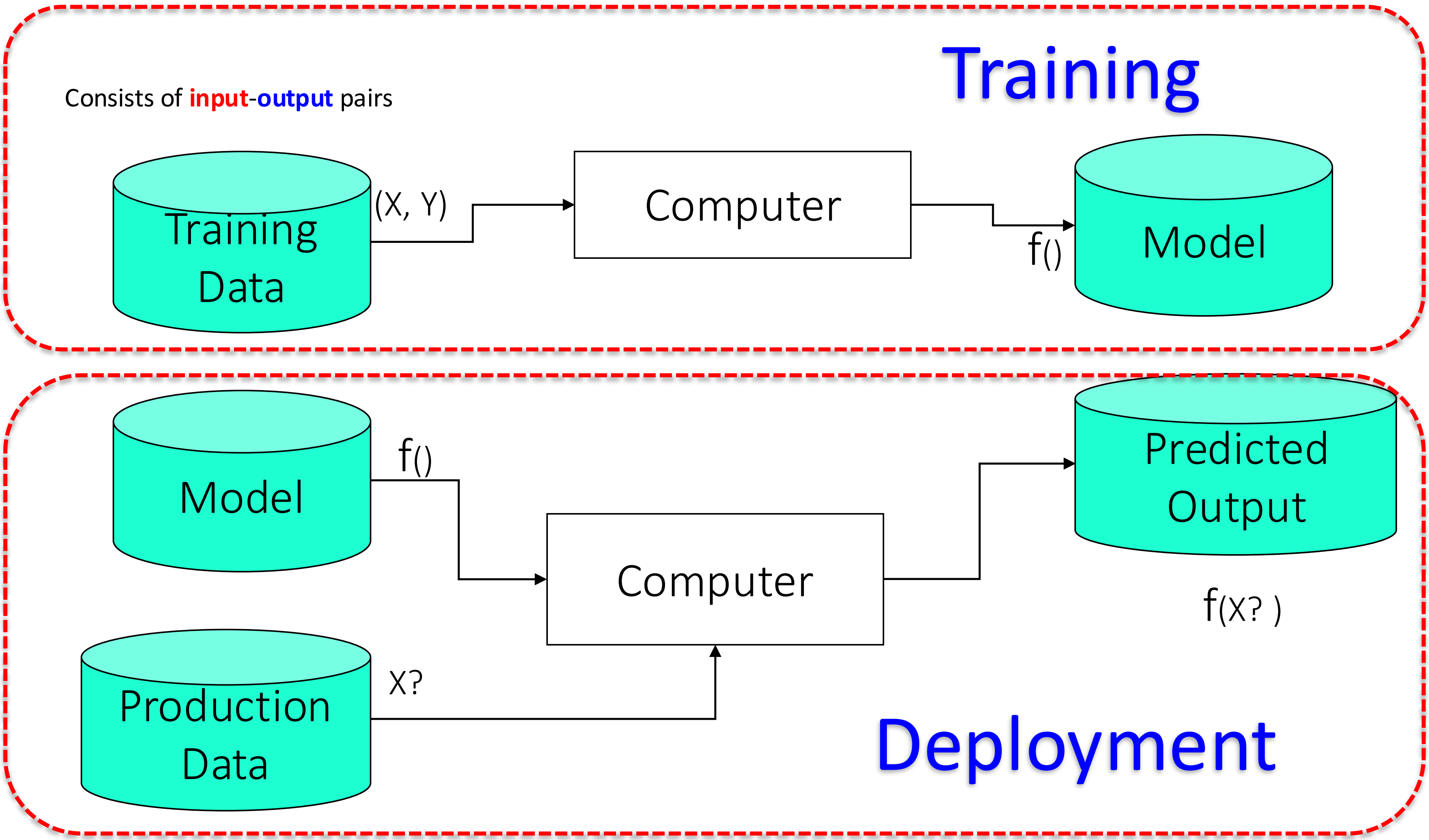
Traditional Programming



Machine Learning



Two Modes of Machine Learning



Machine Learning in a Nutshell

Data

$\updownarrow$

Task

$\updownarrow$

Representation

$\updownarrow$

Score Function

$\updownarrow$

Search/Optimization

$\updownarrow$

Models, Parameters

$\updownarrow$

Hardware

ML grew out of work in AI

Optimize a performance criterion using example data or past experience,

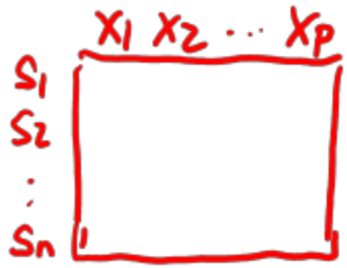
Aiming to generalize to unseen data

Rough Sectioning of this Course

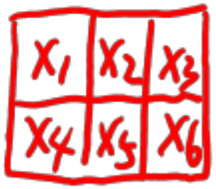
- S1. Basic Supervised Regression + Tabular Data
- S2. Basic Deep Learning + 2D Imaging Data
- S3. Generative and Deep + 1D Sequence Text Data
- S4. Advanced Supervised learning + Tabular Data
- S5. Not Supervised
- S6: Wrap Up + (a few invited tasks, e.g. on AWS)

Course Content Plan ➔ Regarding Data

- ❑ Tabular / Matrix



- ❑ 2D Grid Structured: Imaging

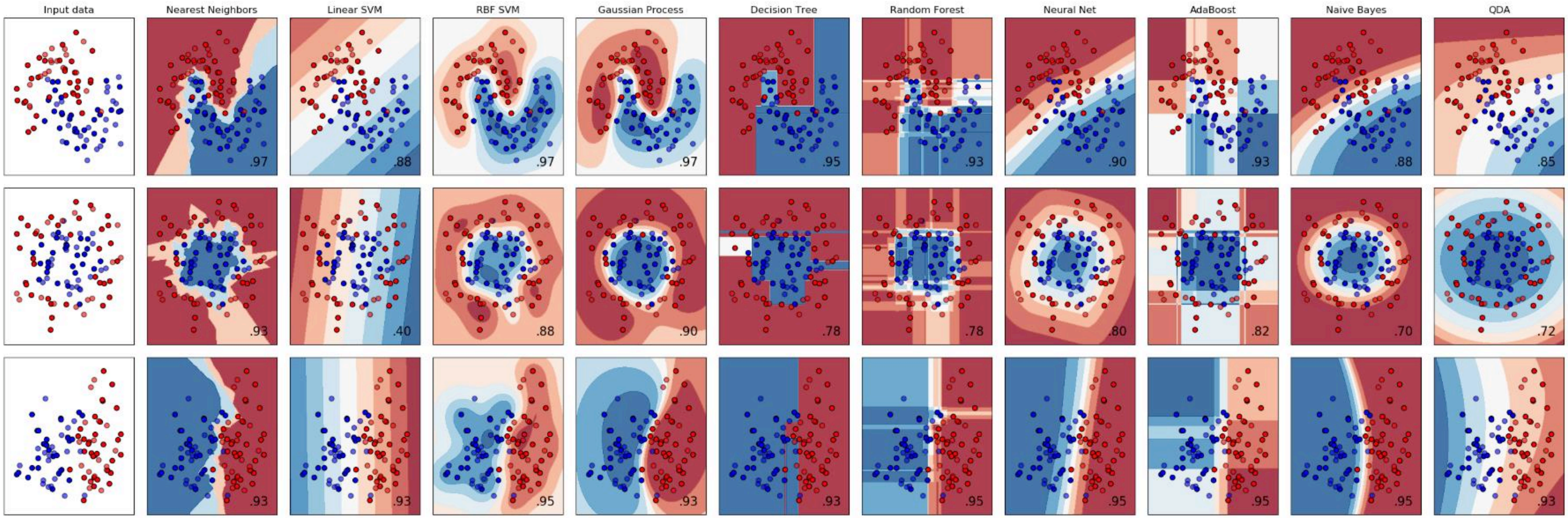


- ❑ 1D Sequential Structured: Text
- ❑ Graph Structured (Relational)
- ❑ Set Structured / 3D /

Course Content Plan → Regarding Tasks

- Regression (supervised) — Y is a continuous
- Learning theory — About f()
- Classification (supervised) — Y is a discrete
- Unsupervised models — NO Y
- Graphical models — About interactions among Y,X1,. Xp
- Reinforcement Learning — Learn to Interact with environment

https://scikit-learn.org/stable/auto_examples/classification/plot_classifier_comparison.html



✓different assumptions on data

✓different scalability profiles at training time

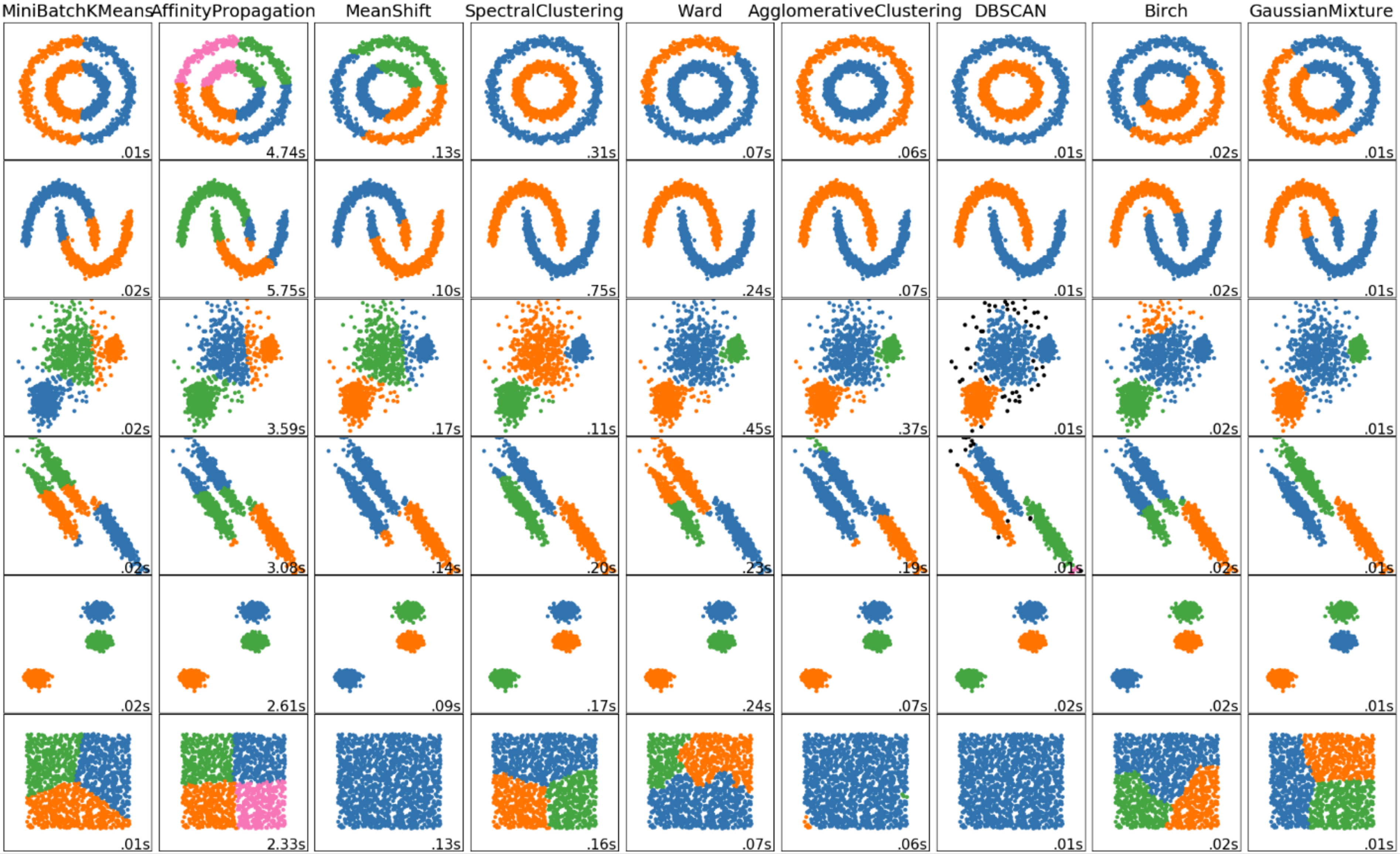
✓different latencies at prediction (test) time

✓different model sizes (embedability in mobile devices)

✓different level of model interpretability / robustness

Adapt from Olivier Grisel's talk

https://scikit-learn.org/stable/auto_examples/cluster/plot_cluster_comparison.html



- ✓different assumptions on data
- ✓different scalability profiles
- ✓different model sizes (embedability in mobile devices)

Quiz 1

Choose correct answers:

Q1: Given the definitions of A and B below, compute AB.

2 points

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ -1 & 0 \\ 5 & 2 \end{bmatrix}$$

Option 1

$$\begin{bmatrix} -5 & -1 \\ 17 & 8 \end{bmatrix}$$

✓

Option 2

$$\begin{bmatrix} -5 & 1 \\ -17 & 8 \end{bmatrix}$$

Option 3

$$\begin{bmatrix} -3 & 1 \\ 15 & 8 \end{bmatrix}$$

Quiz 1

Q2: For conformable matrices A and B, which of the following always holds? 2 points

- (AB)^T=(B^T)(A^T) ✓
- (AB)^T=AB
- (AB)^T=AB^T

Quiz 1

Choose correct answers:

Q3:

2 points

If a matrix $D \in R^{5 \times 7}$, which of the following must always be true?

- Rank(D) = 5
- Rank(D) = 7
- Rank(D) <= 5 ✓
- Rank(D) >= 5
- Rank(D) >= 7

Quiz 2

1. In f:X->Y, which represents the model?

- (•) f
- ( ) X
- ( ) Y

2. How does unsupervised learning differ from supervised learning?

- ( ) No input X is provided
- (•) No label Y is provided
- ( ) Label y is continuous
- ( ) Label y is discrete

Quiz 2

3. Generalization refers to how well your model performs on

- (selected) The testing set
- The training set
- Both the training and testing set

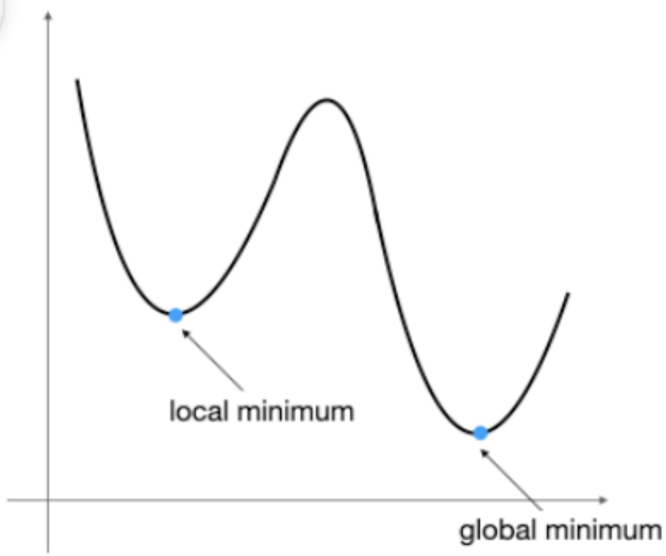
4. The difference between classification and regression is?

- Different types of input
- (selected) Different types of output
- Different types of model
- Different types of programs

Quiz 3 Plus

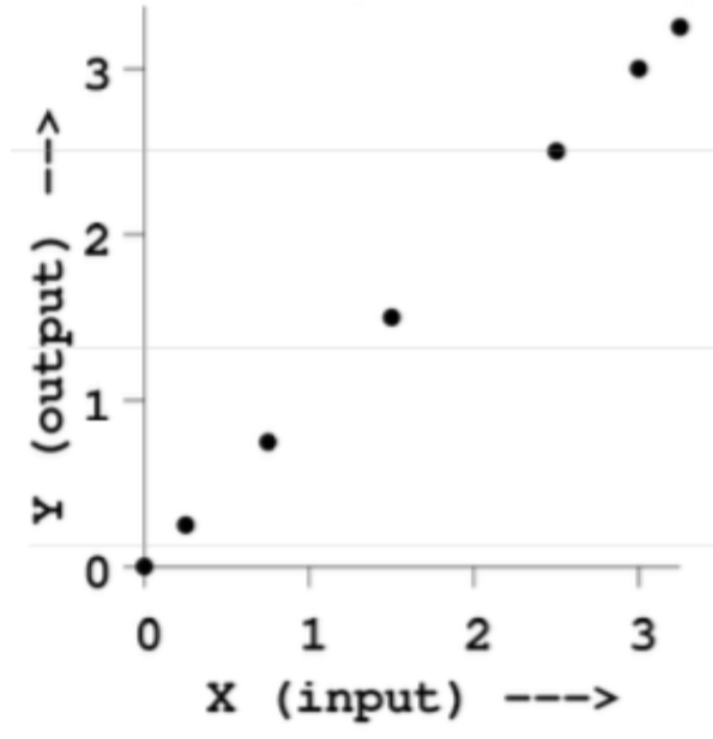
2. True or False? Gradient descent always finds the global minimum. (Hint: Imagine the initial value starts from local minimum, the gradient there is 0)

Multiple choice



- False
- True

Question 3.1. Linear Regression+ Train-Test Split



Quiz 3 Plus

Figure 1: A reference dataset for regression with one real-valued input (x as horizontal axis) and one real-valued output (y as vertical axis).

What is the mean squared training error when running linear regression to fit the data ? (i.e., the model is $y = \beta_0 + \beta_1 x$). Assuming the rightmost three points are in the test set, and the others are in the training set. (you can eyeball the answers.)

Week 3,4- Reading Questions

1. Fundamentals of Linear Regression

- What does it mean for a dataset to be a good fit for linear regression?
- Does linear supervised regression only work with data that is already somewhat linear?
- When is it a good time to use linear regression, and under what conditions will it perform best?
- Where is linear regression used in real-world applications today?
- What challenges exist in making linear regression models robust and trustworthy?
- How should we interpret regression coefficients when features are correlated? Does GD handle multicollinearity?
- What does the “bias” term represent conceptually?

Week 3,4- Reading Questions

- **2. Loss / Cost Functions**
- What exactly is the meaning of Mean Squared Error (MSE), Mean Absolute Error (MAE), Sum of Squared Errors (SSE)?
- When is MSE preferred over MAE, and what are the trade-offs?
- Why is SSE chosen in linear regression instead of MAE?
- What is the purpose of the ½ factor in quadratic loss?
- How do loss functions differ for convex, concave, and saddle point graphs?
- Where did the SSE loss measurement originate from?
- What is the difference between objective, cost, and loss function terminology?
- How do we choose performance metrics (MSE, MAE, R^2, others) and when should multiple metrics be combined?

Week 3,4- Reading Questions

- **3. Gradient Descent & Optimization**
- How does gradient descent (GD) work conceptually?
- What's the difference between GD, stochastic GD (SGD), and mini-batch GD?
- How do we choose learning rate (α) values? Are they fixed or dynamic?
- How do we pick good starting points for GD?
- How does GD behave near local minima, saddle points, or flat regions?
- Are there ways for SGD to escape local minima/saddle points?
- What are good batch sizes, and how do they affect convergence?
- What are the limitations of GD and strategies to overcome them?
- How do we evaluate convergence and know when to stop?
- Could you show a full worked-out example of optimizing with GD step by step?

Week 3,4- Reading Questions

- **4. Normal Equation vs Iterative Methods**
- When should we use the Normal Equation versus Gradient Descent or SGD?
- What are the computational trade-offs between closed-form (Normal Eq.) and iterative (GD/SGD) methods?
- What happens if the feature matrix X does not have full rank?
- Why is Strassen's algorithm for matrix multiplication not always the default, despite being faster in theory?

- **5. Model Selection & Trade-offs**
- How do we know when to choose linear regression vs. more complex models (e.g., Random Forest, SVC)?
- How do we evaluate trade-offs between generalization, efficiency, scalability, and interpretability?
- How does context influence model selection and visualization choices?
- How are these classical regression/optimization topics applied to modern LLMs like ChatGPT or Alexa?

Week 3,4- Reading Questions

- **6. Training, Testing, and Generalization**
- How do we decide the split between training and testing data? (e.g., 80/20 rule)
- Is there an optimal ratio of training to test size?
- What does it mean for a model to generalize well? Does it just mean low error?
- Why is performance on training data not a good indicator of generalization?
- What happens if train/test sets have slightly different distributions (distribution shift)?
- When should validation sets be introduced in addition to train/test splits?
- How does generalization relate to overfitting/underfitting (residual patterns, feature poisoning examples)?

- **7. Matrix & Representation Issues**
- Why represent regression in matrix form? How does it help computation and parallelization?
- What's the difference between summation form and matrix form of the loss function?
- How do row vs. column vectors work in NumPy?
- What does it mean for a matrix to be full rank, and why does it matter?

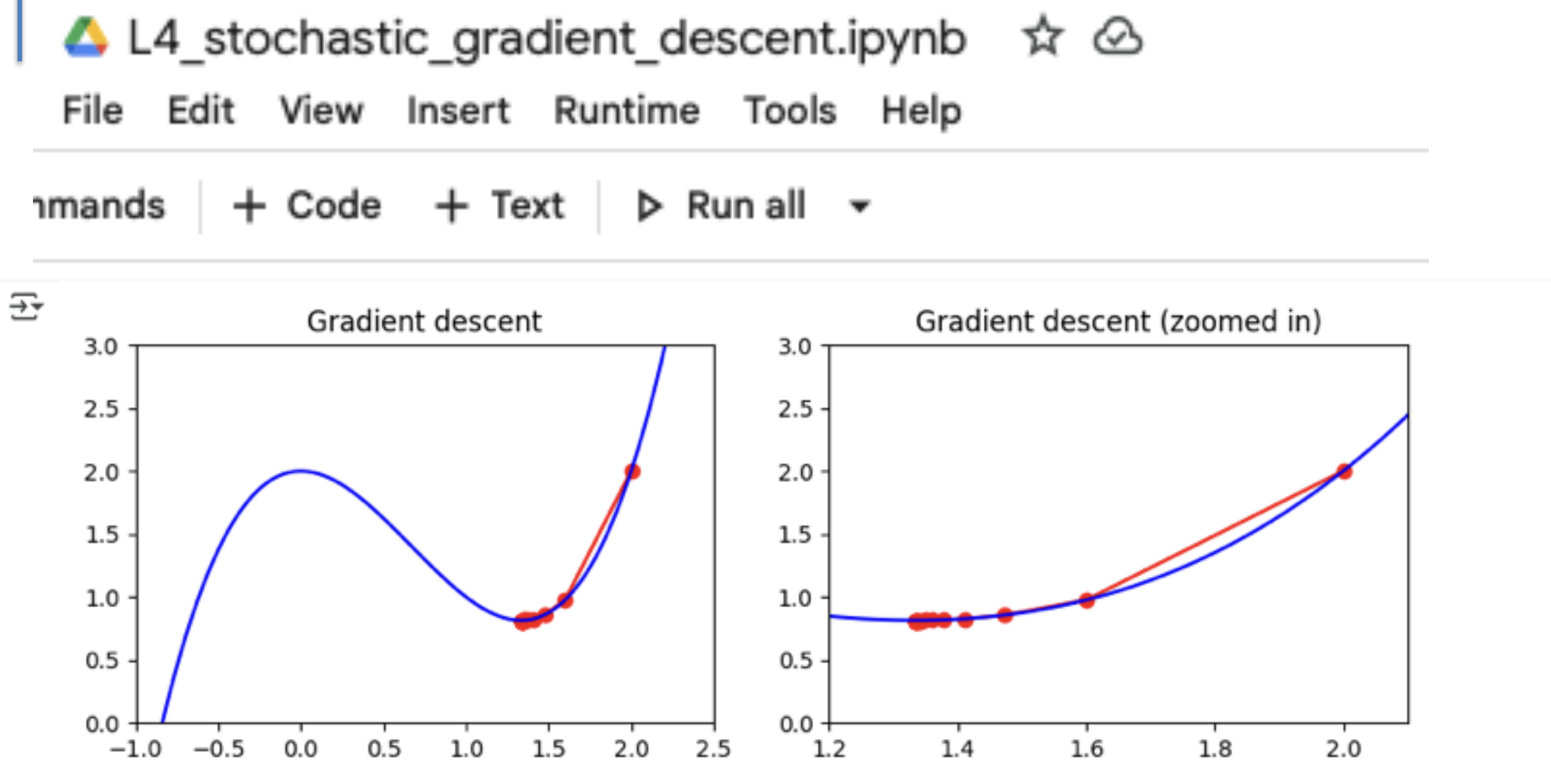
Matrix Representation (p53-)

Lecture 3: Linear Regression Basics

Many architecture details and Algorithm details to consider

- (1): Data parallelization through CPU SIMD / Multithreading/ GPU parallelization / ....
- (2): Memory hierarchical / locality
- (3): Better algorithms, like Strassen's Matrix Multiply and many others

Learning Rate Code Run



Polynomial Regression Code Run

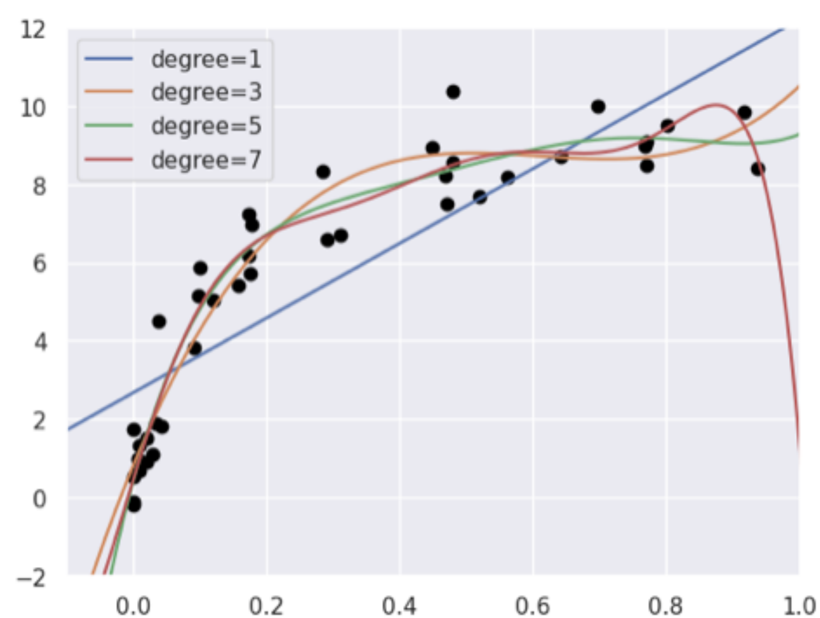
L5-Poly-Regression.ipynb

File Edit View Insert Runtime Tools Help

ımands + Code + Text ▷ Run all

More Regression / Modified from :

1. https://github.com/jakevdp/PythonDataScienceHandbook
2. https://jakevdp.github.io/PythonDataScienceHandbook/05.03-hyperparameters-and-model-validation.html



Quiz 3 Plus

1. True or False: For linear regression, the loss function (sum of squared errors) is convex, meaning gradient descent is guaranteed to find the global minimum if the learning rate is chosen appropriately.

- (•) True
- ( ) False

Quiz 3 Plus

2. In linear regression, what is the role of the intercept term?

- ◯ It scales the input features
- ◉ It shifts the regression line vertically
- ◯ It reduces the variance of predictions
- ◯ It normalizes the input data

Quiz 3 Plus

4. Suppose we want to minimize the function f(w)=w^2+4w using gradient descent. The initial value is w_0=2, and the learning rate is α=0.1. What will be the value of w_1 after one gradient descent update?

- ○ w_1 = 1.6
- ○ w_1 = 2.4
- ○ w_1 = −2
- ◉ w_1 = 1.2

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